

13	D	From $\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT \Rightarrow m = \frac{3kT}{c_{rms}^2} \Rightarrow m \propto \frac{T}{c_{rms}^2}$ $\frac{m_X}{m_Y} = \frac{T_x}{T_y} \left(\frac{c_{rms,Y}}{c_{rms,X}} \right)^2 = \left(\frac{T_x}{2T_x} \right) \left(\frac{3c_{rms,X}}{c_{rms,X}} \right)^2 = \frac{9}{2} = 4.5$
14	B	Both the inlet and outlet temperatures and the room temperature must be kept the same